

DigiPen Institute of Technology

CS 372- CS 596: Assignment # 1

Objectives:

- Show how to use built-in libraries to build K Nearest Neighbor (KNN) and Naïve Bayes classifier.
- Implement KNN from scratch.
- Evaluate KNN & Naïve Bayes Models and analyze the results.
- Compare and analyze according to the results of two algorithms, KNN and Naive Bayes.
- Understand how we apply KNN and Naïve Bayes to real world problems.

Question 1 [44 points]

Note: Use Python to solve this assignment as it has many well-established data science libraries. In this assignment we provided some helpful links to some of the libraries that you will be using.

Instructions

- Download the *Iris Flower* dataset from course Moodle and use it for Questions below.
- Load Data from CSV file [to Panda's data frame](#).

[40 points] Exploratory Data Analysis For Iris Dataset.

1. **[20 points]** Check for missing data and data duplication. Upon detection,
 - a. **[10 points]** show records with missing values.
 - i. **[3 points]** Implement both mean and median imputation.
 - ii. **[4 points]** Compare mean/median imputation vs. dropping rows. Which one is better for sepal_width?
 - b. **[10 points]** Show duplicated records.
 - i. **[4 points]** Explain clearly how you dealt with them. Justify your choice.
 - ii. **[3 points]** Explain why you think we have duplicates (what it means).
2. **[12 points]** Draw [Box Plot](#) to spot outliers. Create boxplots for each feature grouped by species. Make sure to add necessary labels for each plot. Note that you could create Boxplot graphs using [Seaborn](#) library.
 - a. **[2 points]** Explain how Box Plot consider outliers in the dataset.
 - b. **[3 points]** Analyze and comment on the plot.
 - c. **[2 points]** Identify which species has the most outliers and hypothesize why (e.g., measurement errors).
3. **[8 points]** Create [scatter plot matrix](#) for Iris flowers descriptive features. Make sure to add necessary labels to each plot such axes, etc.

- a. **[3 points]** Explain which two features show the clearest separation between species.
 - b. **[3 points]** Explain the plot.
4. **[8 points]** Plot [Heatmap](#) correlation for descriptive features (no species). Make sure to clearly indicate the name of the features on the heatmap as indicated in the dataset and also display/show the value of the correlation on the heatmap.
 - a. Remember that [correlation](#) matrix *helps us to quantify and summarize the **direction and strength** of relationships between variables*. The heatmap shall show:
 - b. **correlation between each pair of descriptive features (X)** (in our case flowers attributes) - we need to know correlation between features and avoid multi-correlation features.
 - i. Clearly indicate positively and negatively correlated features.
 - ii. analyze and comment on the plot.

Split the data

[3 points] In the following parts of question 1, split your data 80/20. Train the model on the 80% fraction and then evaluate the accuracy on the 20% fraction.

1. Make sure that we preserve the proportion of each class in the splits.
2. Make sure that if I run the algorithm again, I will get the same split and also the proportion of the classes are preserved in the split. Remember to shuffle the data before split.

KNN [31 points]

3. **[2 points]** Use [StandardScaler](#) to put all descriptive features on the same scale.
4. **[8 points]** In this part we need to Implement KNN using sklearn library (use [KNeighborsClassifier](#)).
5. **[8 points]** Experiment with different distance metrics, namely, Euclidean metric and Manhattan distance. Compare between them and comment on your findings.
6. **[5 points]** Choose the best number of neighbors using 5-fold cross validation (using [built-in](#)) approach we studied in the class.
7. **[4 points]** K value vs. [Accuracy](#): draw a 2D plot to show the average accuracy of KNN classifier vs different number of k's. Analyze your result and comment on your plot.
8. **[4 points]** Evaluate your model on the test data using Accuracy based on best K found in above. Comment on the result and print misclassified flowers in table format where you show the true label in one column and the predicted table in another column.

Naive Bayes [10 points]

1. **[5 points]** You need to implement Naïve Bayes using the [built-in library](#) and use it to predict the Species of iris in the test dataset.
2. **[2 points]** Use Accuracy to assess the performance of your classifier.

3. **[3 points]** Print misclassified flowers in table format where you show the true label in one column and the predicted table in another column and comment on your output.